

Data Quality Assessment & Preprocessing

Project Overview

This project focuses on assessing data quality and applying common data preprocessing techniques to an e-commerce orders dataset.

Dataset

The dataset used is **Ecommerce_Orders_Missing.csv**. It contains 120 rows and 9 columns, including numerical and categorical features. The dataset also contains missing values and outliers for preprocessing practice.

Tasks Performed

- 1 **Data Quality Assessment:** Checked dataset shape, data types, missing values, and descriptive statistics.
- 2 **Handling Missing Values:** Numerical missing values were filled using the median, while categorical missing values were filled using the mode.
- 3 **Outlier Detection and Handling:** Outliers in Order_Amount were detected using the IQR method and values outside the lower and upper bounds were removed.
- 4 **Feature Scaling:** Applied Min-Max Normalization and Z-score Standardization.
- 5 **Principal Component Analysis (PCA):** PCA was applied to the standardized numerical features. The first two principal components were visualized, and their total explained variance was approximately 43.32%.

Libraries Used

- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn

Files

- **Ecommerce_Orders_Missing.csv** - dataset
- **.ipynb notebook** - preprocessing and analysis code
- **README** - project description

Conclusion

The dataset was successfully cleaned and prepared for further analysis or machine learning. Missing values and outliers were handled, numerical features were scaled, and PCA was used to reduce dimensionality and

analyze the variance captured by the principal components.